

MATH 1342 – Elementary Statistics – Week 6: Probability Rules and Sampling Distributions

Sep 29 – Oct 1 | P.1 Probability Rules • 3.1 Sampling Distributions | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
6. Analyze data presented in two-way tables to provide information about relationships between categorical variables. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions.	1. Compute probabilities for equally likely outcomes and apply the complement rule. (SLO 3) 2. Apply the addition, multiplication, and conditional probability rules using a two-way table, and distinguish disjoint events from independent events. (CO6) 3. Classify a numerical summary as a parameter or a statistic and use the correct notation (μ, σ, p, ρ vs. $\bar{x}$, s, $\hat{p}$, r). (CO8) 4. Describe how a sampling distribution is built from repeated samples, what each dot represents, and its center and shape. (CO8) 5. Explain the standard error as the variability of a statistic and predict how sample size affects it. (CO8)	e-Text: P.1 Probability Rules	MO1, MO2	Participation: Probability – Rock and Roll Hall of Fame two-way table; cards with and without replacement	MO1, MO2	Online Homework: P.1 Probability Rules (due Sun Oct 4, 11:59 PM)	MO1, MO2
		Interactive Lecture: P.1 Probability Rules	MO1, MO2				
		Slides: P.1 Probability Rules	MO1, MO2	Participation: Sampling Distributions – M&M activity: each student draws 10 M&Ms, computes $\hat{p}$ (blue), and adds it to a class dotplot; name the population, sample, and what each dot represents	MO3, MO4, MO5	Online Homework: 3.1 Sampling Distributions (due Sun Oct 4, 11:59 PM)	MO3, MO4, MO5
		e-Text: 3.1 Sampling Distributions	MO3, MO4, MO5				
		Interactive Lecture: 3.1 Sampling Distributions	MO3, MO4, MO5			Take-Home Quiz 3 (due Tue Sep 29, 11:59 PM) – covers 2.5	None
		Slides: 3.1 Sampling Distributions	MO3, MO4, MO5				
						Test 1 – Fri Oct 2 – Sun Oct 4 (Blackboard, proctored, 2 hours). Covers Weeks 1–5.	None